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Structure and Dynamics of [MMIM][Br] Ionic Liquid Confined in Hydrophobic and Hydrophilic Porous matrices: A Molecular Dynamics Simulation Study

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Abstract

Confinement effect on structural and dynamical properties of 1, 3 – dimethylimidazolium bromide ([MMIM][Br]) ionic liquid (IL) have been investigated by molecular dynamics simulation. We use zeolite faujasite (NaY) as hydrophilic confinement and dealuminated faujasite (DAY) as hydrophobic confinement. Presence of extra framework cation $[Na^+]$ in NaY makes the host hydrophilic, whereas DAY with no extra framework cation is hydrophobic. Though both NaY and DAY have almost similar structures, IL shows markedly different structural and dynamical properties in these confinements and in bulk. In confinement, cation-cation radial distribution function which strongly depends on temperature, exhibits layer like structure whereas in bulk it shows liquid like structure which hardly depends on temperature. Though $[MMIM]^+$ and Br^- interaction in DAY is stronger than both NaY and bulk, the strength of interaction between them is almost invariant with temperature. Both $[MMIM]^+$ and Br^- strongly interact with Na^+ of the host and their interaction strongly depends on temperature whereas interaction of IL with Si and O are very small and invariant with temperature. In bulk, self-diffusion coefficient $[D]$ of both $[MMIM]^+$ and Br^- increases exponentially with temperature and D of cation is slightly more than anion at all temperatures we studied whereas in confinement, $[MMIM]^+$ moves much faster than Br^- . For example, in hydrophilic confinement D of cation is 20 – 30 times more than anion. D of both the ions decreases significantly in confinements as compared to the bulk. During diffusion, $[MMIM]^+$ diffuses closer to the inner surface of the hydrophilic confinement than in hydrophobic confinement. Diffusion pathway imperceptibly depends on temperature but strongly depends on the nature of the confinement. Self-part of time dependent Van-Hoove correlation function of $[MMIM]^+$ in hydrophilic confinement shows larger deviation from its Gaussian form than hydrophobic confinement at all temperatures, indicating long-time dynamics of $[MMIM]^+$ in NaY is more heterogeneous than in DAY. Though orientational relaxation timescales of $[MMIM]^+$ in the confinements significantly slowed down as compared to the bulk, confinement does not affect librational motion of the collective hydrogen bond network present in the IL.

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1. Introduction

Room temperature ionic liquids (RTILs) are organic salts exhibiting melting temperature below 100 °C. ILs have strikingly different properties from other molecular liquids such as negligible vapour pressure, non-volatility, high electrical conductivity, non-inflammability melting point, wide range of electrochemical window, thermal stability, etc. Due to their many such interesting properties, researchers are attracted to study comprehensively these class of compounds in the past decade¹⁻⁸. Intrinsic properties of ILs can be adjusted to specific application by the choice of cation-anion combination. As ILs are non-volatile they have favourable solubilising characteristics, it is realised that ILs may be good alternatives to traditional organic solvents^{9, 10}. The growing use of ILs in the field of material science originates from their unique physiochemical and structural properties¹¹. A large number of theoretical and experimental studies on confined ILs have considered carbon and silica matrices¹¹⁻¹³. It is reported that in confinement structural and dynamical properties of ILs change drastically from their bulk properties. Besides carbon and silica matrices, very little is known about ILs confined in chalcogenides¹⁴. The use of ILs in the preparation of wide range of hybrid materials including open-framework structures, silicas, metal oxides, nanostructured metals and alloys have been studied recently¹⁵⁻¹⁷. In these hybrid materials, ILs are confined within a solid matrix known as ionogels¹⁸. The intrinsic hybrid character of ionogels depends on the combinations of solid like host network and ILs. The properties of ionogels are expected to originate from both IL and solid component forming the host matrix. The solid component may be inorganic such as silica and carbon nanotube, organic such as cellulose or organo-inorganic. In general ionogels that retain the properties of ILs, significantly widen their array of applications. Some of these applications ranges from solid electrolyte to drug delivery to catalysis. Confinement of ILs can infer various changes in their physiochemical behaviour due to ion-wall interaction beside ion-ion interaction. First insight regarding effect of confinement is given by thermal behaviour of ILs. Melting point of confined ILs shifts to the lower temperature¹⁹ and this shift is greater than any other molecular liquids²⁰. Lowering in melting temperature is related to the melting temperature of the ILs, pore diameter, enthalpy of fusion, molar volume and surface tension at the solid and liquid-substrate interfaces^{21, 22}. Hence strong dependence of thermal behaviour on the chemical nature of the host is well expected.

It is known that aluminosilicates such as zeolites are good molecular sieves^{23, 24} and it has wide varieties of crystal structures^{25, 26}. They can be used for the synthesis of hybrid materials where ILs are confined within a zeolite and their molecular sieving properties can be tuned. Despite a few studies on the confinement effect on ILs, there is no theoretical/molecular dynamics simulation study on the specific interaction between zeolite and IL. Depending on the structure of host framework, different effect on IL is expected²⁷. Among various kinds of zeolites, faujasite is the most extensively studied zeolite^{28–30} and effect of confinement on different properties of the guest is well explored^{31, 32}. Presence of extra framework cation [Na⁺] in the faujasite structure is known as zeolite NaY and it is hydrophilic. Replacement of Al with Si atom in faujasite produces a crystal structure similar to the crystal structure of zeolite NaY with the exception that it does not contain any Na⁺ ions. This dealuminated faujasite zeolite is known as DAY and it is hydrophobic. Though both NaY and DAY have almost similar structures, different effect of confinement is expected due to hydrophilic and hydrophobic nature of the host network. Hence, it is important to understand the effect of different type of confinements on thermal, structural and dynamical behaviour of RTILs.

2. Simulation Methods

All atom molecular dynamic (MD) simulations are performed to study different properties of [MMIM][Br] IL in bulk and in hydrophobic and hydrophilic confinement by using DL_PLOY³³. $2 \times 2 \times 2$ unit cell of NaY and DAY in a cubic simulation cell of length of 49.7 Å and 48.7 Å respectively have been chosen as the host network in all simulations. To study bulk properties of [MMIM][Br], total number of ion pairs is fixed to 100 in all the simulations. Figure 1 (a) shows that density of IL slightly decreases with temperature. As the unit cell volume of NaY and DAY zeolite does not change within the temperature we studied, total number of ion pairs inside the confinements is varied with temperature to maintain the bulk density of IL. All simulations are performed at five different temperatures, namely 400 K, 450 K, 500 K, 550 K and 600 K. The total number of ion pairs in $2 \times 2 \times 2$ unit cells at these temperatures are 152, 150, 146, 142 and 132 respectively. The structure of [MMIM][Br] is shown in Figure 1 (b).

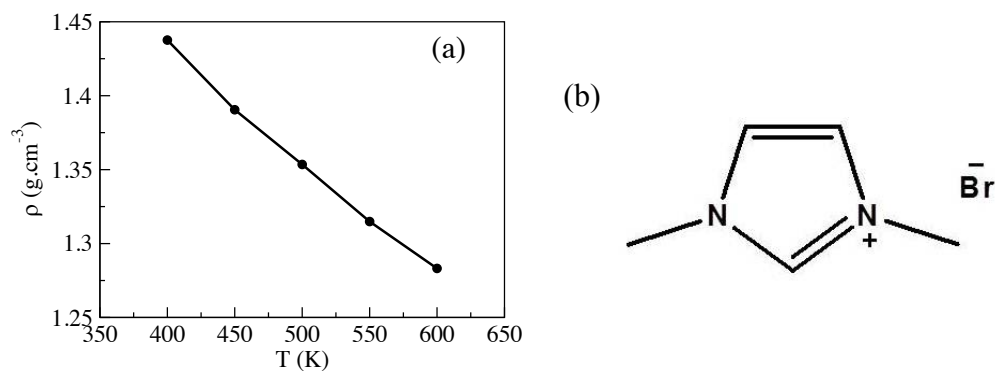


Figure 1. (a) Variation of density of [MMIM][Br] with temperature and (b) structure of [MMIM][Br] ionic liquid

In this work [MMIM]⁺ is considered to be flexible and interacting via OPLS-AMBER type force field with the potential form

$$E = \sum_{bonds} K_r (r - r_0)^2 + \sum_{angles} K_\theta (\theta - \theta_0)^2 + \sum_{dihedrals} \frac{K_\phi}{2} [1 + \cos(n\phi - \delta)] + \sum_{i < j} \left[\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} + \frac{q_i q_j}{r_{ij}} \right]$$

where first three terms represent harmonic bond stretching, angle bending and dihedral torsion respectively. Final term in the parenthesis represents the short range 12-6 Lennard-Jones (LJ) potential and long range Coulomb potential between two interacting atoms i and j . All the force field parameters for [MMIM]⁺ and [Br]⁻ are taken from literature.^{34, 35} In this study, Zeolite NaY and DAY are considered to be flexible using the force field of Aurbach³⁰. LJ parameters for the cross interaction terms are computed by using Lorentz-Berthelot combination rule³⁶. For all the simulations, we start with a random configuration of ILs and then energy minimization is done by using conjugant gradient methods which eliminates all bad contacts in the starting configuration. After energy minimization, we heat the system to 700 K at constant volume which ensures proper demixing throughout the host network and then subsequent cooling to our desired temperature with step-down process with a step of 50 K. In each step we equilibrate the system for 2 ns in NVT ensemble for the confinements and for bulk the equilibration is done in NPT ensemble. In all NPT

simulations, we use Noose- Hoover thermostat and barostat with coupling constants 0.5 ps and 2 ps respectively^{37, 38}. This temperature annealing procedure ensures proper mixing of IL and it is very essential for the attainment of equilibrium. Once the system reaches at the desired temperature, we equilibrate all the systems for 20 ns in NVT ensemble. To analyse different properties of the systems, production run of 20 ns is performed in all cases. Noose-Hoover thermostat with a coupling constant of 0.5 ps is used in this study. Long range electrostatic interactions are tackled with Ewald summation method³⁶ with a real space cut off of 16 Å. Equation of motion is integrated by using velocity verlet algorithm³⁶ with a time step of 1.0 fs. The cut-off distance for non-bonded interaction is chosen to be 16 Å. All properties are computed from the trajectories stored at an interval of 1.0 ps during the production run.

3. Results and Discussions

a. Structural properties

Radial distribution function

We compute radial distribution functions (rdfs), $g(r)$ between different components of the system to understand the effect of confinement on structural property of the IL. Rdfs also provide molecular level understanding regarding the strength of interaction between different ions. We assign centre of imidazolium ring of the cation as CR and regard this as the “head” of the cation. Figure 2 (a), (b) and (c) show head – head rdfs in hydrophilic and hydrophobic confinements and in bulk respectively at different temperatures. In bulk, nature of rdfs are very different as compared to that in hydrophilic and hydrophobic confinements. We do not observe any sharp peak for neat IL, only a broad peak which is seen at 7.2 Å is invariant with temperature, whereas a small peak which is seen at 4.0 Å, disappears above 450 K. In both hydrophilic and hydrophobic confinements a sharp peak is seen at 3.7 Å and it's intensity decreases significantly with temperature. The peak intensity in DAY is more than NaY, particularly at 400 K and 450 K suggesting closer approach of the [MMIM]⁺ ions among themselves due to hydrophobic effect of the host matrix. We also observe an intense second peak at 8.3 Å in hydrophobic confinement whereas for hydrophilic confinement the second peak is at 7.4 Å. In DAY, intensity of the second peak is invariant with temperature but in DAY, the peak intensity decreases with significant broadening.

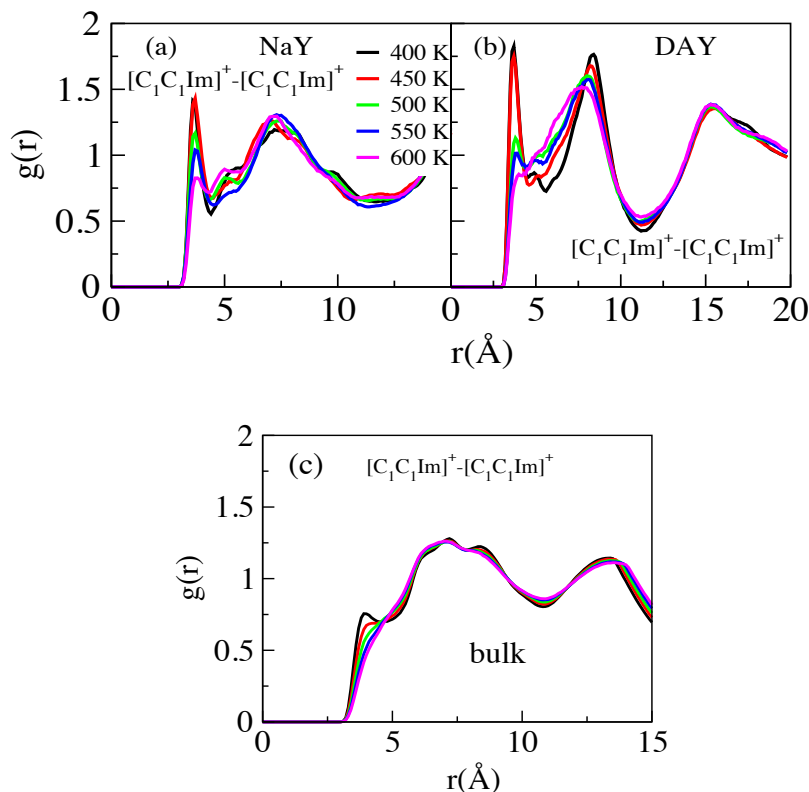


Figure 2. Head - head radial distribution functions (rdfs) at different temperatures in (a) NaY, (b) DAY and (c) bulk.

The appearance of intense peaks in cation - cation rdfs in the confinements suggest significant confinement induce increase of cation - cation interaction. The existence of layer like structure due to strong cation - cation interaction have already been observed in other simulation studies on IL at the solid interfaces^{39,40}. Figure 3 (a), (b) and (c) show $Br^- - Br^-$ rdfs in the confinements and in bulk respectively at different temperatures. The distributions in confinements are markedly different as compared to bulk. In both hydrophilic and hydrophobic confinements, Br^- shows layered like structure whereas in bulk it has liquid like structure. The strength of anion - anion interaction in confinements is more stronger than in bulk, particularly in NaY it is ~3 times more than in bulk.

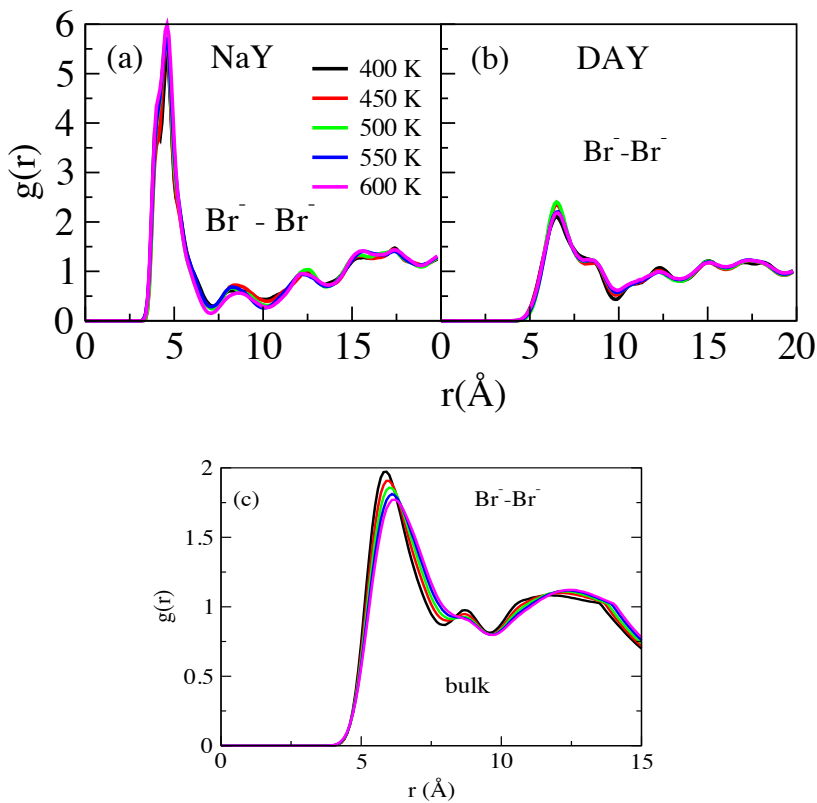


Figure 3. Anion - anion radial distribution functions (rdfs) at different temperatures in (a) NaY, (b) DAY and (c) bulk.

Figure 4 (a), (b) and (c) show $[\text{MMIM}]^+ - \text{Br}^-$ rdfs in hydrophilic and hydrophobic confinements and in bulk respectively. Though nature of CR - anion rdfs in bulk and in confinements are nearly similar, the first peak height in confinements is more than in bulk, suggesting stronger cation - anion interaction in confinements as compared to bulk. In all cases, except imperceptible decrease in intensity of the first peak, there is no change in cation - anion rdfs as a function of temperature. This signifies that cation - anion near neighbour correlation does not depend significantly on temperature. Another striking difference between the two confinements is that cation - anion interaction strength in hydrophobic zeolite is always higher than hydrophilic zeolite and neat IL.

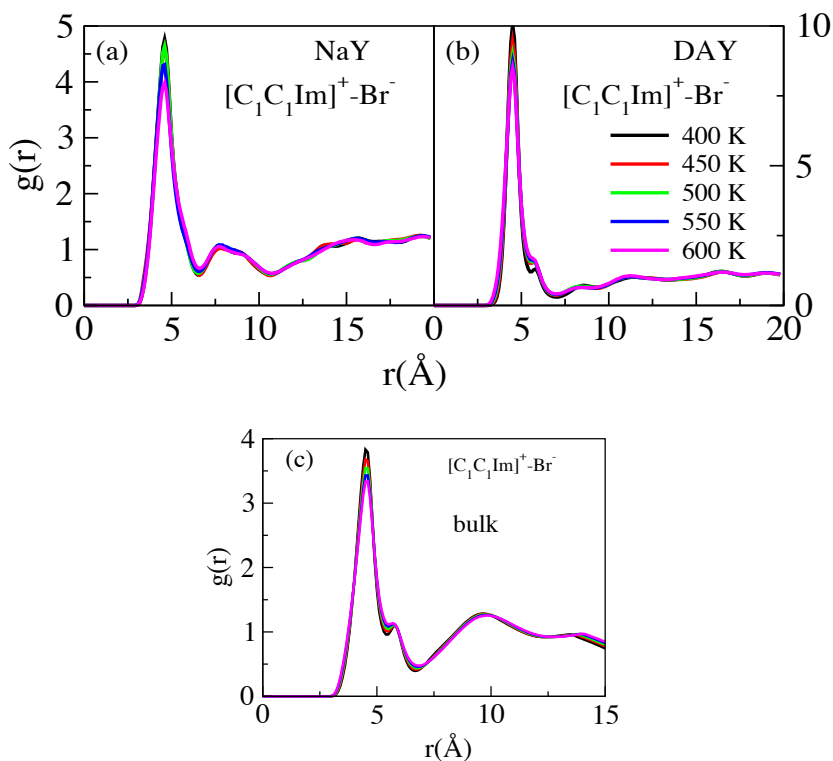


Figure 4. Cation-anion radial distribution functions at different temperatures in (a) NaY, (b) DAY and (c) bulk.

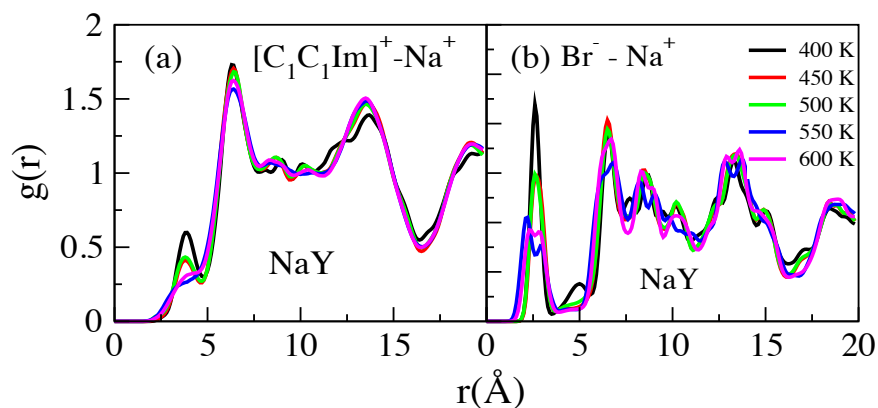


Figure 5. Radial distribution functions at different temperatures in NaY between (a) cation- Na^+ (b) anion- Na^+ .

Figure 5 (a) and (b) show rdfs between $[MMIM]^+$ and Na^+ of the host NaY and $Br^- - Na^+$ respectively. As expected, we observe high affinity of Br^- towards Na^+ and presence of multiple

sharp peaks indicate layered structure of Br^- around extra framework Na^+ . Though first peak height in $\text{Br}^- - \text{Na}^+$ rdfs significantly reduces with temperature, layering phenomenon between them exists even at 600 K. Interaction between $[\text{MMIM}]^+$ and Na^+ is very weak as indicated by a small peak at $r = 3 \text{ \AA}$ in Figure 5 (a). The nature of $[\text{MMIM}]^+ - \text{Na}^+$ rdfs does not alter significantly with temperature except the small peak at $r = 3 \text{ \AA}$ disappears at high temperature.

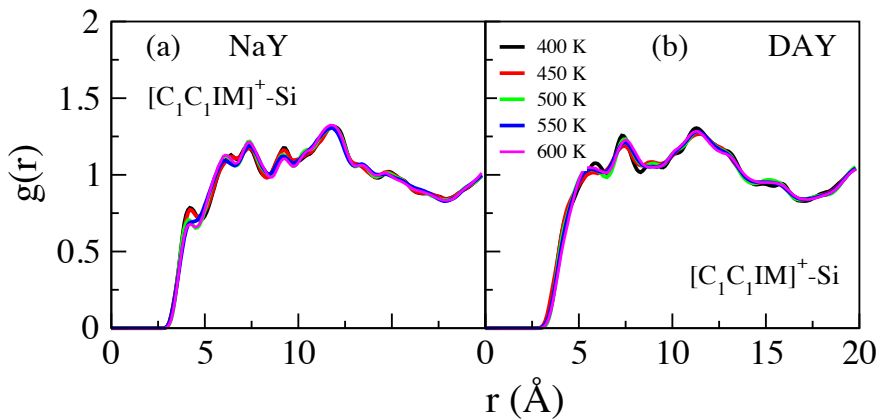


Figure 6. Radial distribution functions of $[\text{MMIM}]^+$ and Si in (a) NaY and (b) DAY at different temperatures.

Figure 6 (a) and (b) show temperature dependent rdfs of $[\text{MMIM}]^+$ and Si in NaY and DAY respectively. For both confinements, we observe multiple small peaks with the first peak at $4.2 \text{ \AA}$ and $5.3 \text{ \AA}$ in NaY and DAY respectively. As temperature increases, first peak height in NaY slightly decreases and at 550 K it completely disappears whereas for hydrophobic confinement rdf is invariant with temperature. Figure 7 (a) and (b) show $[\text{MMIM}]^+ - \text{O}$ rdfs at different temperatures in hydrophobic and hydrophilic confinements respectively. They show similar temperature dependence as discussed in Figure 6.

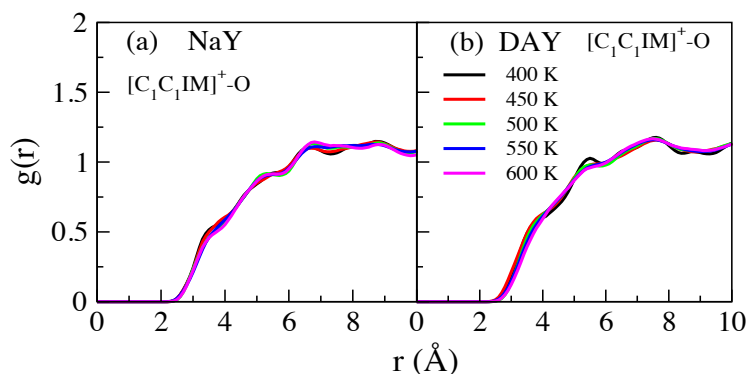


Figure 7. Radial distribution functions at different temperatures between $[\text{MMIM}]^+$ and O in (a) NaY and (b) DAY.

b. Dynamical Properties

Diffusion coefficient

To understand the influence of confinement on translational motion of IL, self-diffusion coefficient is computed from the mean square displacement (MSD) by using Einstein's relation⁴¹

$$D = \frac{1}{6t} \langle |\Delta r(t)|^2 \rangle$$

MSD is computed from the c. o. m. positional vectors (r_i) of each ion from the relation given below

$$\langle |\Delta r(t)|^2 \rangle = \frac{1}{N} \left\langle \sum_{i=1}^N |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \right\rangle$$

where $\mathbf{r}_i(t)$ is the position of a particular ion at time t . It is known that Lopes force field under predicts the self-diffusion coefficient of both cation and anion in bulk and it is one order of magnitude less than the experiment though considering polarization effect it gives better agreement with the experiment^{32,42}. In bulk as well as in confinements, simulated $\langle |\Delta r(t)|^2 \rangle$ shows multiple time dependences where $\langle |\Delta r(t)|^2 \rangle \propto t^p$ with $p = 1$ (diffusion dominated long time behaviour), $p = 2$ (inertia dominated short time behaviour) and $p < 1$ (sub-diffusive behaviour at intermediate time scales). Temperature dependence MSDs show substantial increase in sub-diffusive time region in confinement than in bulk (see supporting information). In bulk, sub-diffusive time regime for cation is ~ 100 ps to 10 ps whereas it is ~ 200 ps to 15 ps in hydrophobic

confinement and ~ 250 ps to 20 ps in hydrophilic confinement in the temperature range 400 K to 600 K. In bulk, sub-diffusive time regime for anion is $\sim 240 - 24$ ps, it is $\sim 250 - 170$ ps in hydrophobic confinement and $\sim 825 - 300$ ps in hydrophilic confinement. Though sub-diffusive time regime of anion is much more than cation in bulk as well as in confinements, the difference in cation and anion sub-diffusive time scales is much higher in the confinements as compared to neat IL. Upon rise in temperature, substantial decrease in sub-diffusive time scale in confinements and in bulk indicates occurrence of thermal energy induced cage breaking and more facile movement of the ions. Figure 8 shows D , computed from the slope of linear fit to the MSDs of cation and anion within the time span of 1.0 ns to 2.5 ns at various temperatures in confinements and in bulk. Earlier it was reported that in bulk self-diffusion coefficient of cation is slightly higher than anion although cation has larger size than anion^{43,44}. It was observed that as temperature increases, difference in D between cation and anion increases. We also observe that in bulk, D of both $[\text{MMIM}]^+$ and Br^- increases exponentially with temperature and D of cation is always higher than anion. For example, D of $[\text{MMIM}]^+$ is 1.2 times higher than Br^- at 600 K.

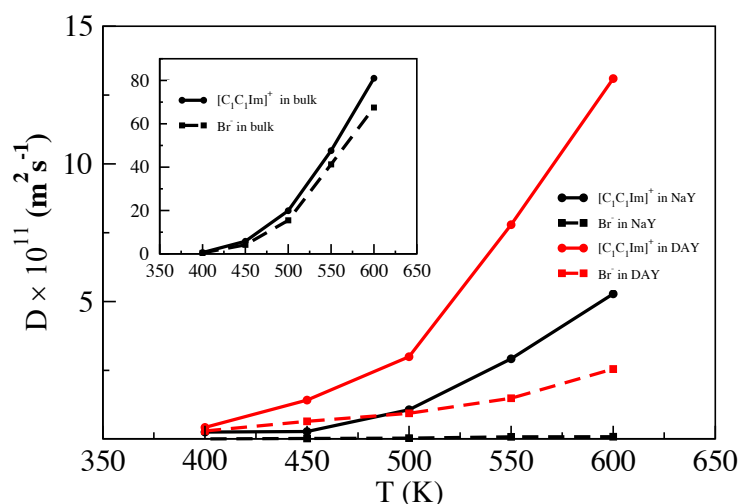


Figure 8. Self-diffusion coefficient of $[\text{MMIM}]^+$ and Br^- in two different confinements at various temperatures. Inset figure shows D of cation and anion of neat IL as a function of temperature.

Due to confinements, D of $[\text{MMIM}]^+$ and Br^- decreases significantly as compared to bulk and cation moves much faster than anion. At 600 K, D of $[\text{MMIM}]^+$ in hydrophilic confinement is 16

times slower than bulk whereas D of Br^- is two orders of magnitude slower as compared to bulk. At the same temperature, D of $[\text{MMIM}]^+$ in hydrophobic confinement is 6 times slower than bulk and Br^- is 26 times slower as compared to bulk. Hence, hydrophilic confinement has much more effect on IL translation motion than hydrophobic confinement. Figure 5 (a) and 6 (a) show a small peak at 3.8 Å and 4.2 Å respectively in $[\text{MMIM}]^+ - \text{Na}^+$ and $[\text{MMIM}]^+ - \text{Si}$ rdfs. These peaks were absent in hydrophobic confinement. In hydrophilic confinement, multiple peaks in Figure 6 (b) indicates strong affinity of Br^- towards Na^+ . Absence of extra framework Na^+ in DAY is responsible for facile movement of the anion in hydrophobic confinement. Due to strong interaction of IL with the host NaY, decrease in D in hydrophilic confinement is more than in DAY. Another striking feature is that the difference in D of cation and anion is very large in confinements as compared to bulk. At 600 K, D of cation is 42 times higher than anion in hydrophilic confinement whereas in hydrophobic confinement D of cation is only 5 times higher than anion. It is also observed that unlike bulk where D of both cation and anion increases significantly, in the confinements D of cation increases more rapidly than anion and increase in NaY is much higher than in DAY as temperature increases. This is because the first peak height in $[\text{MMIM}]^+ - \text{Si}$ decreases with temperature whereas in DAY, density distribution is almost invariant with temperature (see Figure 6). Due to strong $\text{Na}^+ - \text{Br}^-$ interaction, increase in D with temperature is almost negligible in hydrophilic confinement.

From the Arrhenius plot ($\ln D$ vs $1/T$) (see supporting information for the linear fit to the Arrhenius plot), we compute activation energy of $[\text{MMIM}]^+$ and $[\text{Br}]^-$ in bulk and in the confinements shown in Table 1. Interestingly, we observe that in both the confinements activation energy of $[\text{MMIM}]^+$ is higher than Br^- though D of $[\text{MMIM}]^+$ is much more than Br^- , particularly at high temperature. In bulk, activation energy of both cation and anion is nearly equal and much more as compared to both the confinements but D of the ions in neat IL is much higher than in the confinements. Instead of activation energy, strong guest – host interaction plays a crucial role on long time diffusion in the confinement. Recently, it is observed that activation energy of water inside AOT reverse micelle is less than that in bulk⁴⁵.

Table 1. Activation energy of [MMIM]⁺ and [Br]⁻ in bulk and in two different confinements.

Ions	NaY (kJ/mol)	DAY (kJ/mol)	bulk (kJ/mol)
[MMIM] ⁺	33.2	35.0	50.3
[Br] ⁻	22.7	20.9	52.1

Probable diffusion pathway of cation in the confinements

In order to find out the diffusion pathway of cation during cage to cage intercage migration, we compute distribution of [MMIM]⁺ at different planes perpendicular to the line joining window center and corresponding α -cage center. First we compute the time steps when a guest crosses from one α -cage to another α -cage through the connecting window. Then we consider 4 ps before and after intercage migration to find out location of the diffusing species and then bin all those locations to different planes of 0.5 Å width. The distance of diffusing species in a particular plane from its center are designated as R and the distance between the cage center and window center is chosen as r (Please see Fig. S11 in supporting information for the details). The plane at $r = 0$ represents the window plane and as r increases, it moves towards the α -cage center. Location of diffusing species in a particular plane is governed by guest-guest as well as guest-host interaction. In a particular plane, R_{max} is the point where probability of finding a diffusing species is maximum and then connecting all these R_{max} in different planes through a line gives the most probable intercage diffusion path. Figure 9 (a), (b) and (c) show the most probable diffusion path of [MMIM]⁺ in two different confinements at 400 K, 500 K and 600 K respectively. In hydrophilic confinement, [MMIM]⁺ diffuses closer to the inner surface than hydrophobic confinement. Diffusion pathway imperceptibly depends on temperature but strongly depends on the nature of the confinement.

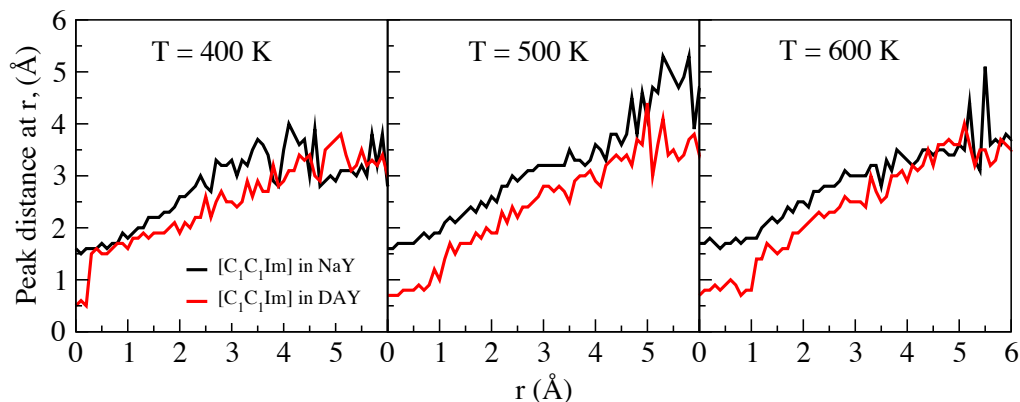


Figure 9. Most probable diffusion pathway of $[\text{MMIM}]^+$ in two different confinements at (a) 400 K (b) 500 K and (c) 600 K.

Dynamical heterogeneity

As it is known that pure ILs are not only structurally heterogeneous but also dynamically heterogeneous^{43, 46}, we study effect of different confinements on dynamical heterogeneity at various temperatures. Generally presence of Dynamical Heterogeneity (DH) in a system is quantified through non-Gaussian parameter, $\alpha_2(t)$ ^{47, 48} and the self-part of van Hove correlation function, $G_s(\mathbf{r}, t)$ ⁴⁹. Non-Gaussian parameter is defined as

$$\alpha_2(t) = \frac{3 \langle |\Delta \mathbf{r}(t)|^4 \rangle}{5 \langle |\Delta \mathbf{r}(t)|^2 \rangle^2} - 1$$

where $\Delta \mathbf{r}(t)$ is the displacement of a particular ion at time interval t and can be computed from MSD. $\alpha_2(t)$ is zero at $t = 0$ and passes through a maximum at some intermediate time interval t before it again goes to zero at $t \rightarrow \infty$. Self-part of the time dependent van Hove correlation function is defined as

$$G_s(\mathbf{r}, t) = \frac{1}{N} \left\langle \sum_{i=1}^N \delta(\mathbf{r}_i(t) - \mathbf{r}_i(0) - \mathbf{r}) \right\rangle$$

where N is the number of particular ions present in the system. $G_s(\mathbf{r}, t)$ measures the probability of a molecule i moves from its position $\mathbf{r}_i(0)$ at time zero to position $\mathbf{r}_i(t)$ at time t . For simple liquids, $G_s(\mathbf{r}, t)$ is always Gaussian in normal condition. However for supercooled liquid it deviates from the Gaussian form.^{49, 50} For dynamically heterogeneous system, $\alpha_2(t)$ is non-zero and the self-part of van Hove correlation function deviates from its Gaussian behaviour. Figure 9 (a), (b) and (c) show $\alpha_2(t)$ for [MMIM]⁺ in two different confinements and in bulk respectively at various temperatures. $\alpha_2(t)$ shows monotonic temperature dependence with peaks at different time scale (τ_{NG}). Like other ILs⁴³, $\alpha_2(\tau_{NG})$ for neat [MMIM][Br] decreases with temperature (see Figure 10 (c)). In both the confinements, $\alpha_2(\tau_{NG})$ shows similar temperature dependence like in bulk but at any temperature $\alpha_2(\tau_{NG})$ is much higher in confinements than in bulk. In case of hydrophilic confinement, $\alpha_2(\tau_{NG})$ are 2.9, 2.7, 2.6, 2.1 and 1.7 and in hydrophobic confinement they are 2.7, 2.0, 1.5, 1.2 and 1.0 at 400 K, 450 K, 500 K, 550 K and 600 K. Interestingly, at all temperatures $\alpha_2(\tau_{NG})$ of cation is more in NaY than DAY, suggesting translational motion of the cations are less heterogeneous in hydrophobic confinement than hydrophilic confinement. At short times, $\alpha_2(t)$ does not depend on time or confinement because in the initial temporal regime, the dynamics are ballistic at all temperatures. This is indicated by the first shoulder in Figure 10 (a) and (b). Table 2 lists τ_{NG} of [MMIM]⁺ in hydrophilic and hydrophobic confinements and in bulk at long time for different temperatures. It is seen that τ_{NG} is minimum in bulk, maximum in hydrophilic confinement and it takes intermediate value in hydrophobic confinement at any temperature. The dynamical features of cation can also be characterised by self-term of $G_s(\mathbf{r}, t)$. We compute $G_s(\mathbf{r}, t)$ at three different temperatures in NaY and DAY at $t = \tau_{NG}$ and compared with the predicted Gaussian behaviour, $G_g(\mathbf{r}, t)$. $G_s(\mathbf{r}, t)$ and $G_g(\mathbf{r}, t)$ are shown in Figure 11 (a), (b) and (c) respectively at 400 K, 500 K and 600 K. Cation in hydrophilic confinement shows larger deviation from its Gaussian behaviour than in hydrophobic confinement, indicating temporal heterogeneity associated with [MMIM]⁺ is more in NaY than DAY⁵¹⁻⁵⁴.

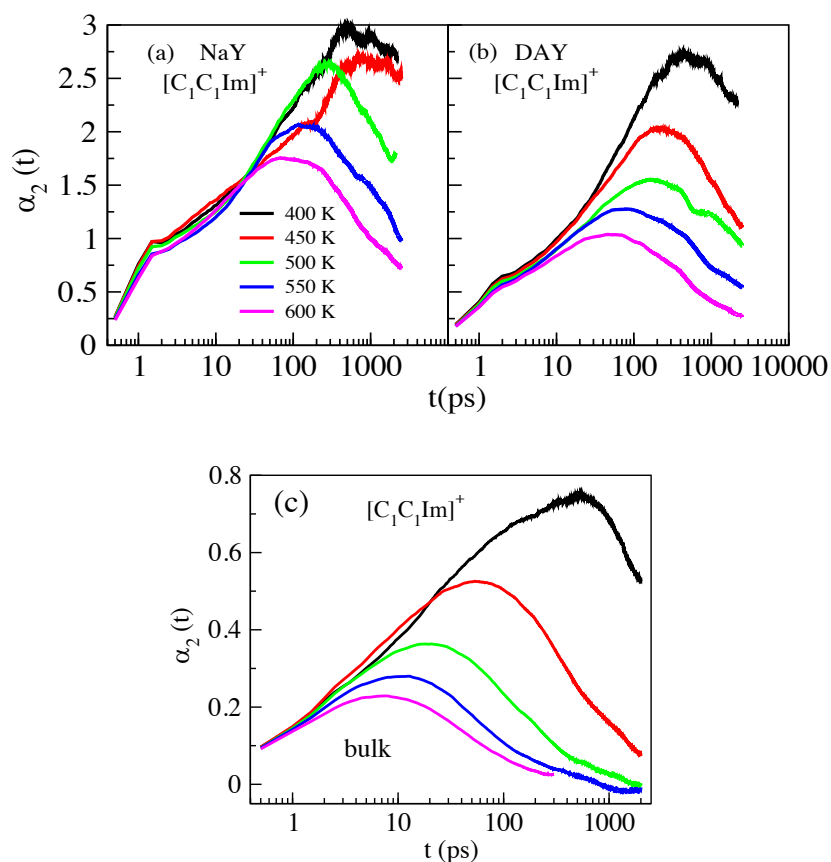


Figure 10. Variation of non-Gaussian parameter, $\alpha_2(t)$ of $[\text{MMIM}]^+$ in (a) NaY, (b) DAY and (c) in bulk at different temperatures.

Table 2. Dependence of τ_{NG} of $[\text{MMIM}]^+$ at different temperatures in two different confinements.

Temperature (K)	τ_{NG} (ns) (NaY)	τ_{NG} (ns) (DAY)	τ_{NG} (ns) (bulk)
400	0.52	0.51	0.50
450	0.57	0.22	0.06
500	0.28	0.17	0.02
550	0.13	0.08	0.015
600	0.075	0.054	0.007

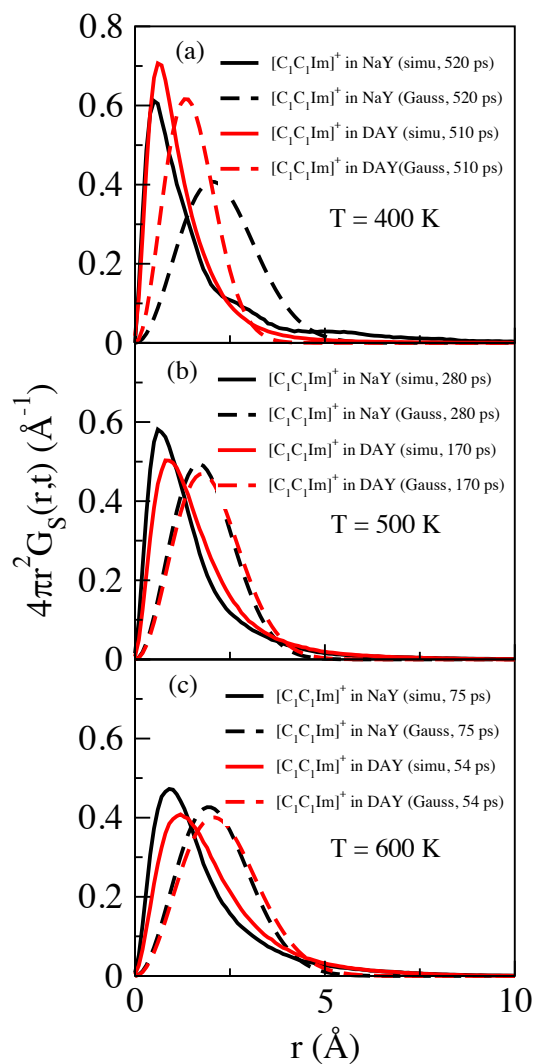


Figure 11. Self-part of the Van-Hoove correlation function along with its Gaussian counterpart for $[\text{MMIM}]^+$ in two different confinements at (a) 400 K (b) 500 K and (c) 600 K.

Reorientational time correlation function

Molecular rotational motion in liquids are usually characterized through reorientational correlation function, $C_l(t)$ defined as

$$C_l(t) = \frac{\langle P_l[\mathbf{u}_i(t) \cdot \mathbf{u}_i(0)] \rangle}{\langle P_l[\mathbf{u}_i(0) \cdot \mathbf{u}_i(0)] \rangle}$$

where P_l is the Legendre polynomial of rank l , $\mathbf{u}_i$ is the unit vector along a given molecular direction about which $C_l(t)$ is computed and angular brackets denote averaging over molecules

and time-origins. In this study, we consider the vector which is perpendicular to the plane of imidazolium ring as the reference vector. Average reorientational time, $\langle \tau_l \rangle$ is defined as the average time taken by a molecule to rotate one radian with respect to its reference vector and is obtained through time integral of the $C_l(t)$ as $\langle \tau_l \rangle = \int_0^\infty C_l(t) dt^{41}$. $l = 1$ is associated with Optical Kerr Effect (OKE), while $l = 2$ is associated with fluorescence anisotropy. So computation of $\langle \tau_l \rangle$ gives direct correlation between simulation and experiment.

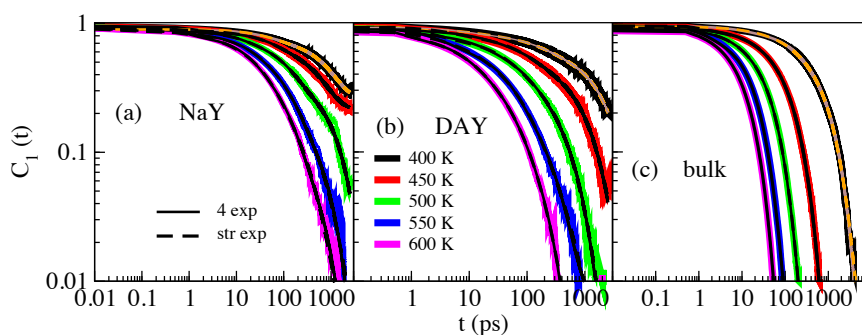


Figure 12. Log-log plot of reorientational time correlation functions (RTCF's) along with the fits for $l = 1$ in (a) hydrophilic confinement, (b) hydrophobic confinement and (c) in bulk respectively at different temperatures.

Figure 12 (a), (b) and (c) represent simulated decays of $C_l(t)$ of cation at five different temperatures in hydrophilic and hydrophobic confinements and in bulk. Although in confinements decays at low temperatures are not complete, yet they provide a qualitative idea regarding orientational relaxation of $[\text{MMIM}]^+$. Clearly, the orientational relaxation in NaY and DAY is not only significantly slowed down as compared to the bulk but also their relaxation patterns are different^{55,56}. For example, at 450 K the decay of $C_l(t)$ at 450 K is complete within 750 ps whereas in hydrophobic and hydrophilic confinements at 2.5 ns, $C_l(t)$ decays upto 95 percent and 80 percent respectively. Figure 13 (a), (b) and (c) show orientational relaxation of $C_2(t)$ in the confinements and in bulk respectively. Like $C_l(t)$, relaxation of $C_2(t)$ in confinements is significantly slowed down. For example, at 450 K the decay of $C_2(t)$ is complete within 740 ps whereas in hydrophobic and hydrophilic confinements at 2.5 ns, $C_l(t)$ decays upto 60 percent and 55 percent respectively.

Interaction between $[\text{MMIM}]^+$ and the host can significantly slowed down the relaxation rates of confined IL. In hydrophilic confinements, due to strong $[\text{MMIM}]^+ - \text{Na}^+$ interaction, decay of both $C_1(t)$ and $C_2(t)$ are slower than hydrophobic confinement. In Figure 12 and Figure 13 both multi-exponential and stretched exponential fits to the IL data have been shown which suggest neither of these functions perform better over the other. All fit parameters and normal plot (without log-log plot) of $C_1(t)$ and $C_2(t)$ are given in the supporting information.

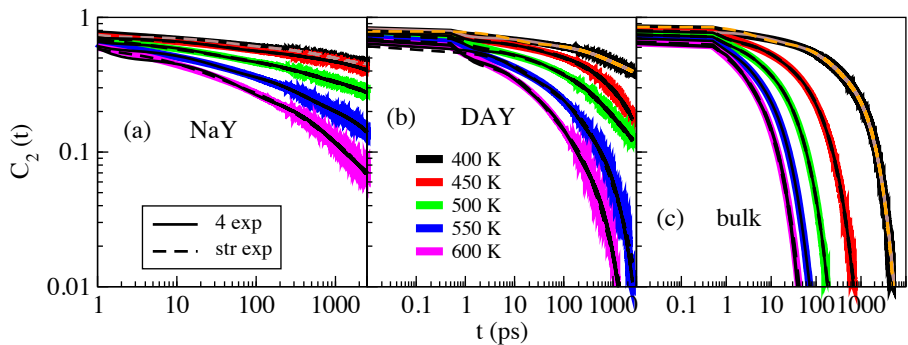


Figure 13. Log – log plot of reorientational time correlation functions (RTCF's) along with the fits for $l = 2$ in (a) hydrophilic confinement, (b) hydrophobic confinement and (c) bulk respectively at different temperatures.

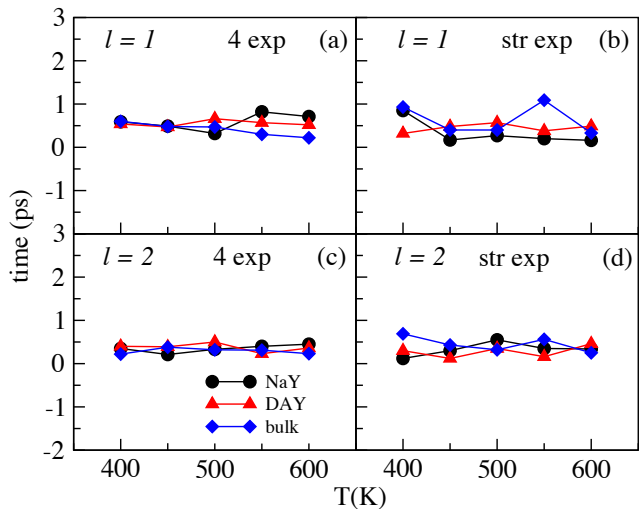


Figure 14. Ultrafast component of the decay timescales for $l = 1$ obtained from (a) multi- and (a) stretched - exponential fit to the data shown in Figure 12 and Figure 13. (c) and (d) are the decay timescales for $l = 2$.

Figure 14 (a) and (b) show ultrafast component of the decay timescales which is obtained from both multi- and stretched- exponential fit to the data shown in Figure 12 (for $l = 1$) in the confinements and in bulk. The fastest relaxation time of $[\text{MMIM}]^+$ in both the confinements and bulk are within ~ 0.5 ps to 1.0 ps, suggesting even in confinements there is no change of the fastest relaxation time. Figure 14 (c) and (d) show the ultrafast component of the decay for $l = 2$ and there is no significant change in the timescales. These suggest that confinement does not affect the librational motion of the collective hydrogen bond network present in the IL.

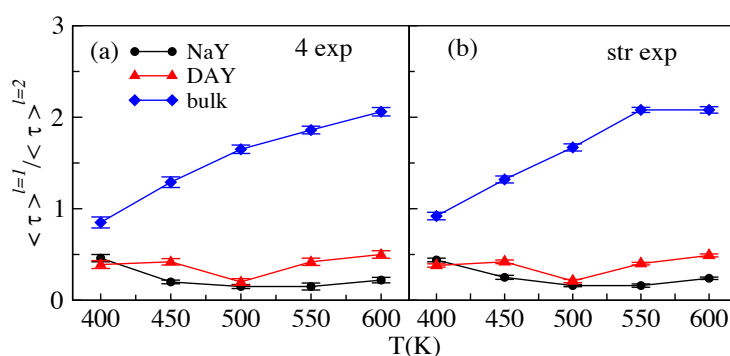


Figure 15. The ratios between the simulated average orientational relaxation timescales ($\langle \tau \rangle$) obtained from both (a) multi-exponential fit and (a) stretched exponential fit to the data shown in Figure 12 and Figure 13.

Let us compare average simulated reorientational relaxation times ($\langle \tau \rangle$) for $l=1$ and 2. Figure 15 (a) and (b) show $\langle \tau \rangle$ obtained from multi-exponential and stretched exponential fit to the data shown in Figure 12 respectively. The average relaxation time denotes amplitude-weighted time constants associated with each of the component. The ratios between the simulated average times, $\langle \tau \rangle^{l=1} / \langle \tau \rangle^{l=2}$ for $[\text{MMIM}]^+$ in bulk are in the range of ~ 1 to 2 which is much deviated from 3 predicted on the basis of stochastic Brownian dipolar rotations⁵⁴. This reflects presence of temporal heterogeneity of neat IL at all temperatures we studied. It is interesting to note that in the confinements the ratio of these rank dependent average times are in the range of ~ 0.25 to 0.51 which is much smaller as compared to the bulk. In the confinements, ratio of $\langle \tau \rangle^{l=1}$ and $\langle \tau \rangle^{l=2}$ is less than one suggest that average relaxation time associated with $l = 2$ is much slower than $l =$

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1 (see supporting information for the detailed timescales), suggesting temporal heterogeneity of [MMIM]⁺ in confinement is more as compared to bulk. It is quite interesting to notice that temporal heterogeneity associated with [MMIM]⁺ is almost comparable in both hydrophilic and hydrophobic confinements as shown by the $\langle \tau \rangle^{l=1}$ and $\langle \tau \rangle^{l=2}$ ratios in Figure 15.

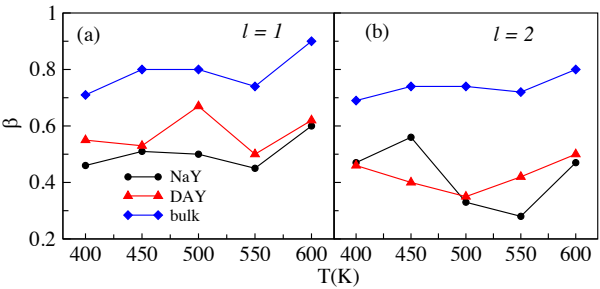


Figure 16. The fit parameter, β obtained from the stretched-exponential fit for (a) $l = 1$ and (b) $l = 2$ respectively to the data shown in Figure 12 and Figure 13.

Figure 16 shows the fit parameter, β obtained from the stretched-exponential fit for $l = 1$ and 2 to the data shown in Figure 12 and Figure 13 (see supporting information for all the fit parameters). The β values in bulk are always less than one and in both the confinements it is lower than bulk, suggesting existence of temporal heterogeneity in these systems which we are also observed from the multi-exponential fit discussed above. The guest-host interaction induces more heterogeneity in confined IL.

Conclusions

Through molecular dynamic simulations in canonical ensemble, we have studied structural and dynamical properties of [MMIM][Br] ionic liquid confined in hydrophobic and hydrophilic faujasite type zeolites and compared the results with neat IL. Though both NaY and DAY have almost similar structures, IL shows markedly different structural and dynamical properties in these confinements and in bulk. Though [MMIM]⁺ and Br⁻ interaction in DAY is always stronger than in NaY and bulk, the strength of interaction between them is almost invariant with temperature. Both [MMIM]⁺ and Br⁻ strongly interact with Na⁺ of the host and their interaction strongly depends on temperature whereas interaction of IL with Si and O are very small and invariant with

temperature. It is observed that with increase in temperature, self-diffusion coefficient of both cation and anion increases significantly in the confinements but unlike bulk, increase in D of $[\text{MMIM}]^+$ is much higher than Br^- . Hence as compared to neat IL, the difference of D for cation and anion increases with temperature. In hydrophilic confinement, $[\text{MMIM}]^+$ diffuses closer to the inner surface of the host than hydrophobic confinement but very close to the zeolite window, preference of intercage crossing remains almost identical irrespective of the confinement and temperature. Self-part of time dependent Van-Hoove correlation function of $[\text{MMIM}]^+$ in hydrophilic confinement shows larger deviation from its Gaussian form than hydrophilic confinement at all temperature studied, indicating long-time dynamics of $[\text{MMIM}]^+$ in NaY is more heterogeneous than DAY. Though orientational relaxation of cation in the confinements significantly slowed down as compared to the bulk, confinement does not affect librational motion of the collective hydrogen bond network present in the IL. Due to guest-host interaction, the temporal heterogeneity associated with $[\text{MMIM}]^+$ is more than neat IL.

Supporting Information

MSDs, log-log plot of MSDs, Arrhenius plot, reorientational correlation function and fit parameters.

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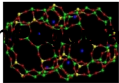
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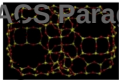
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Two α -cages of zeolite NaY



Two α -cages of zeolite DAY

Physical Chemistry

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